

Key Methodology — Technical Explainer

Black Media Intelligence Pipeline | July 2026

1. Threshold-Tuning Leak

The hybrid tagger has a sensitivity dial — a threshold — that controls how aggressively it tags articles as belonging to a given topic. In our original evaluation, we were tuning this dial on the same 400 articles we used to measure performance. This is equivalent to setting the answer key after seeing the test, and it inflated our results by approximately 0.07 in F1 score.

The fix was a cross-annotator validation protocol: we tune the threshold on Annotator 1's 200 articles and test on Annotator 3's 200 articles — and vice versa. Because the threshold is never set on the same data it is evaluated against, the results are honest and generalise to unseen articles.

Impact: Hybrid Soft Boost F1 dropped from 0.336 (inflated) to 0.267 (honest) after the fix — confirming the leak was real and significant.

2. Inter-Annotator Agreement (IAA)

Before trusting our machine results, we needed to establish how hard this task is for humans. Adeniyi independently re-labeled all 400 articles and compared his labels against each original annotator using Cohen's Kappa — a chance-corrected agreement score where 1.0 is perfect agreement and 0.0 is no better than random.

This gives us a human agreement ceiling: the maximum performance any automated system can realistically be expected to achieve, given that even two trained human annotators working from the same guidelines do not always agree.

Result: Macro Kappa 0.33 (Adeniyi vs Annotator 1) and 0.38 (Adeniyi vs Annotator 3). The task is inherently moderately ambiguous — a known property of social-conflict topic labeling.

3. Supervised Classifier

Instead of hand-crafted rules (keyword matching) or fixed similarity thresholds (semantic tagger), the supervised classifier learns directly from the 400 human labels. For each of the 12

social-conflict topics, a separate logistic regression model is trained using three signals per article: the full article text converted into a 384-dimensional numerical representation of meaning (via the all-MiniLM-L6-v2 sentence embedding model), the keyword tagger's binary prediction for that topic, and the semantic tagger's raw similarity score.

The same cross-annotator validation protocol used for the hybrid tagger is applied here: train on one annotator's articles, test on the other's. Topics with fewer than two positive training examples fall back to the keyword prediction, and this is logged transparently in the evaluation report.

Result: Pooled Macro F1 0.322, Kappa 0.297 — the best performing method across all six approaches evaluated, reaching 77–91% of the human agreement ceiling. This is the method we recommend adopting as the production tagger.